Hierarchical Algorithms

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→ Oral Exam

20 - 25 mins.

I. Introduction & Rest of the topics.

1) What are objectives/goals of Hierarchical matrices

2) What is the underlying idea of Hierarchical matrices to reach the complexity (like $O(n \log n)$)?

product of two matrices with few rows & columns.

3) Arithmetic of rank k matrices

II. 4) We have a dense matrix and we want to reduce it to Hierarchical. How do we do those?

The different methods and questions about those methods.

5) Admissibility condition in interpolation

6) What are bounding boxes in admissibity condition?

Why do we use it?

7) Explain ACA (Adaptive cross approximation)

↳ preferably explain on paper.

8) Compare Tensor interpolation and ACA

with advantages and disadvantages.

Which would you choose and why?

9) Arithmetic of Hierarchical matrices

Complexity of Matrix Vector multiplication

Proof of the Theorem

Sparsity constant?

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10) What is the underlying idea of H^2 matrix compared to H matrix?

11) Complexity of H^2 matrix
we need to explain in detail
like what is stored and how

12) Matrix vector multiplication in exercise

13) Numerical aspects from exercise:
Like what is n, d
What has to be added to get to H^2 from H matrix?
What is n_{min}?
What is δ and θ?
↓ used ACA upper bound for error
└ parameter in CA
diameter / distance

14) Data structure of matrix shown and ask questions about it

15) Questions based on performance

16) Recursive nature of matrix

Questions from all chapters will be tried to covered

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